Naive Lie Theory 3.6 Exercises

OblivionIsTheName

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Notes

I am going to use some topological argument about torus in 3.6.2. A compact, connected, abelian Lie group is torus, and torus is preserved by the image of continuous homomorphism. However, the formal proof of the former statement requires materials (exponential map) covered in later chapter. I will refer back to here then and now only prove the latter.

First recall these concepts. Let X be a group and topological space.

1. Compactness means every open cover has a finite subcover. An open cover of X is a set of open subsets U_α where $G = \bigcup_{\alpha \in A} U_\alpha$; and a subcover is just a subset of the cover.

2. Connectedness means there are no two disjoint open set can cover X.

3. Being abelian means X is commutative.

Let $f : G \to H$ be a continuous homomorphism, and G be compact, connected, and abelian.

1. Now we prove that $f(G)$ is compact. Let C be an open cover of $f(G)$. $\{f^{-1}(U) | U \in C\}$ is an open cover of G. By compactness of G, take a finite subcover $\{f^{-1}(U_i) | i \leq n\}$, and $\{U_i | i \leq n\}$ is a finite cover $f(G)$, and $f(G)$ is compact.

2. Now we prove $f(G)$ is connected. We prove the contrapositive. If $f(G)$ is disconnected, i.e., it is a union of two disjoint open subsets U, V. Therefore $f^{-1}(U)$ and $f^{-1}(V)$ disconnect X.

3. Abelian property is naturally preserved under homomorphism.

3.6.1

Consider the identification map $\varphi : \mathbb{C}^n \to \mathbb{R}^{2n}$ where $\varphi(z_1, ..., z_n) = (\mathrm{Re}(z_1), \mathrm{Im}(z_1), ..., \mathrm{Re}(z_n), \mathrm{Im}(z_n))$. This is clearly a bijection. It's easy to check that $Z_{\theta_1,...,\theta_n} = \varphi^{-1} R_{\theta_1,...,\theta_n} \varphi$.

3.6.2

First we want to embed $\mathrm{U}(n)$ into $\mathrm{SO}(2n)$ by $\rho(A) = \varphi A \varphi^{-1}$. It's an injective and continuous homomorphism. It's easy to check that $\rho(\mathbb{T}^n) \cong \mathbb{T}^n$. Specifically,

$$\rho(Z_{\theta_1,\ldots,\theta_n}) = R_{\theta_1,\ldots,\theta_n}$$

Given $\mathbb{T}^n$ is indeed a torus in $\mathrm{U}(n)$, assume a bigger torus $S \in \mathrm{U}(n)$ with $\mathbb{T}^n \subset S$. Since ρ is injective, inclusions stay strict as

$$\rho(\mathbb{T}^n) \subset \rho(S) \subseteq \mathrm{SO}(2n)$$

And given ρ is continuous homomorphism, so $\rho(S)$ is a torus in $\mathrm{SO}(2n)$ bigger than $\rho(\mathbb{T}^n)$. Contradict.

3.6.3

The proof of maximal tori in the chapter already implies this problem. It shows if A commutes with everyone else, A is in the corresponding maximal torus. The biggest abelian group is a subset of maximal torus, and maximal torus itself is abelian, so maximal torus is in fact the biggest abelian subgroup of these rotation groups.

3.6.4

$\mathrm{O}(n)$ is disconnected. It has $\mathrm{SO}(n)$ and those with -1 determinant. However, any torus is connected and contains identity. Therefore any torus in $\mathrm{O}(n)$ is in $\mathrm{SO}(n)$. On the other hand, obviously every torus in $\mathrm{SO}(n)$ is also in $\mathrm{O}(n)$. So the maximal tori of them are the same.